MAT1375, Classwork10, Fall2026

Ch10. Rational Functions I

1. Definition of the **Rational function**:

A rational function is a fraction of two polynomials $f(x)=\frac{p(x)}{g(x)}$, where $p(x)$ and $g(x)$ are both polynomials, and $g(x)\neq 0$.

The **domain of a rational function** f is all real numbers for which the denominator $g(x)$ is not zero:

$$D_f=\{x\in\mathbb{R} \mid x\in D_p\cap D_g \text{, but } g(x)\neq 0\}$$

2. **Arrow Notation**: Given a constant a and we have

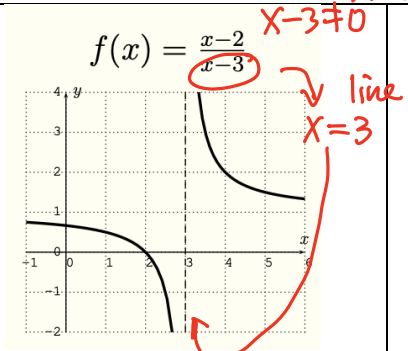
$x\to a^+$:	x approaches a from the **right** (x is very closed to a but $x\neq a$ and $x > a$)
$x\to a^-$:	x approaches a from the **left** (x is very closed to a but $x\neq a$ and $x < a$)
$x\to\infty$:	x approaches infinity (x increases without bound)
$x\to-\infty$:	x approaches negative infinity (x decreases without bound)

3. The definition of a **Vertical Asymptote**:

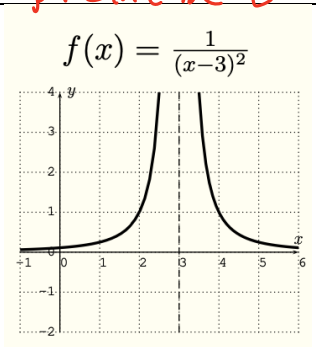
The line $\boldsymbol{x=a}$ is a Vertical Asymptote of the graph of a function f if $f(x)$ **increases or decreases without bound** as $\boldsymbol{x}$ **approaches** $\boldsymbol{a}$. Here are three examples:

x as an input can't be 3

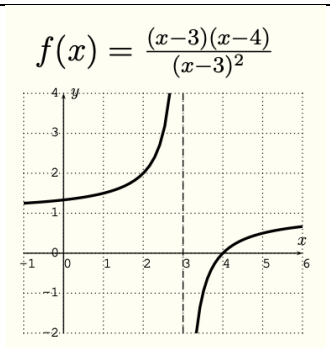
$x-3\neq 0$



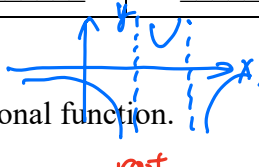
As $x\to 3^+$, $f(x)\to\infty$;
As $x\to 3^-$, $f(x)\to-\infty$.



As $x\to 3^+$, $f(x)\to\infty$
As $x\to 3^-$, $f(x)\to\infty$



As $x\to 3^+$, $f(x)\to-\infty$
As $x\to 3^-$, $f(x)\to\infty$

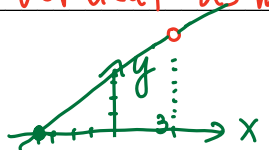


$\frac{(x-3)(x+4)}{(x-3)(x-5)}$

4. How to locate Vertical Asymptotes: Let $f(x)=\frac{p(x)}{g(x)}$ be a rational function.

(1) If $p(x)$ and $g(x)$ have no common factors, and $\boldsymbol{a}$ is a **zero** of $g(x)$ which makes $f(x)$ undefined, then the line $x=a$ is a vertical asymptote of the graph of $f(x)$.

(2) If $\boldsymbol{a}$ is a **zero** of both $p(x)$ and $g(x)$ ($p(a)=0$, $g(a)=0$.) which means $x-a$ is the common factor of $p(x)$ and $g(x)$, then there is a jump / removable discontinuity / singularity at $x=a$ and there is no vertical asymptote at $x=a$.



$y=\frac{(x-3)(x+4)}{(x-3)}=(x+4)$ assume $x-3\neq 0$ input $x\neq 3$ with $x\neq 3$

5. Find the vertical asymptotes of the graph of each rational function:

a) $f(x) = \frac{x}{x^2-1}$ b) $g(x) = \frac{x-1}{x^2-1}$ c) $h(x) = \frac{x-1}{x^2+1}$

a) $f(x) = \frac{x}{(x+1)(x-1)}$ $p(x)=x$, $q(x)=(x+1)(x-1)$, p, q has no common factor(s)

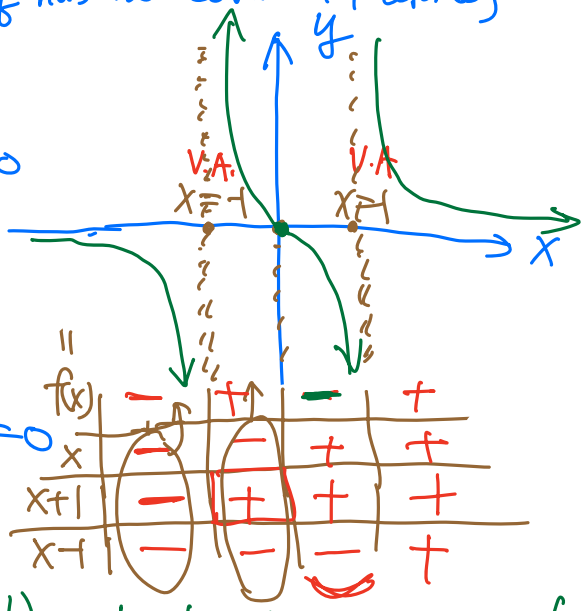
To find vertical asymptotes, let $q(x)=0$

$q(x)=(x+1)(x-1)=0 \Rightarrow x+1=0$ or $x-1=0$

It means that the line $x=-1$ and $x=1$ are the vertical asymptotes (V.A)

A root of $f(x)$ as an input will make $f(x)=0$

$x=0$ is a root of $f(x)$



b) $f(x) = \frac{x+1}{(x+1)(x-1)}$, $p(x)=(x+1)$, $q(x)=(x+1)(x-1)$ and they have common factor $(x-1)$

$= \frac{1}{x+1}$ with $x \neq 1$

when $x=0$, $y=\frac{1}{0+1}=1$, y-intercept

To find vertical asymptotes, it means $x+1=0$

it implies the line $x=-1$ is a vertical asymptote.

